Standard Cell Library Design and Characterization (SKY130)

Project Report

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Chapter 1

Introduction

This project focuses on the design and characterization of a standard cell library using the SKY130 process and NGSPICE. The goal is to generate Non-Linear Delay Model (NLDM) timing tables for a set of logic gates under specified operating conditions. These tables are widely used in digital design flows for accurate timing analysis.

The work involves transistor-level design, functional verification, automated characterization using Python scripts, and analysis of delay behavior under varying input slew and load conditions.

Chapter 2

Methodology

2.1 Technology Setup

All simulations were performed using the SKY130 open-source PDK at:

- Process corner: TT
- Supply voltage: 1.8V
- Temperature: $25°C$

NGSPICE was used as the simulation engine.

2.2 Cell Design

A total of 13 standard cells were implemented:

- Inverters: invx1, invx2, invx4, invx8
- NAND2: nand2x1, nand2x2, nand2x4
- NOR2: nor2x1, nor2x2, nor2x4
- MAJ3: maj3x1, maj3x2, maj3x4

Each cell was implemented at transistor level using SKY130 device models.

2.3 Characterization Setup

Characterization was performed using:

Input transition (ns):

$$0.0100,\ 0.0231,\ 0.0531,\ 0.1225,\ 0.2823,\ 0.6507,\ 1.5000$$

Load capacitance (pF):

$$0.0005,\ 0.0013,\ 0.0035,\ 0.0094,\ 0.0249,\ 0.0662,\ 0.1758$$

Four timing tables were generated:

- cell_rise
- cell_fall
- rise_transition
- fall_transition

Chapter 3

Results

3.1 Inverter Plot

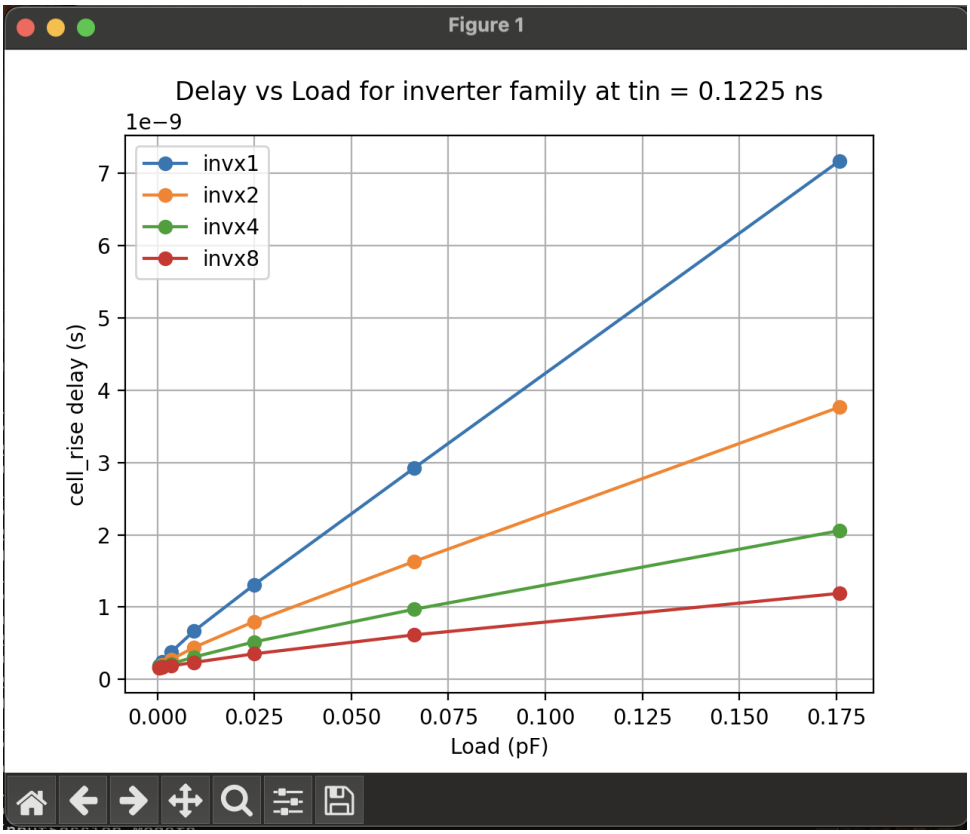

Figure 3.1: Delay vs Load for inverter family at $t_{in} = 0.1225ns$

The plot shows that delay increases with load and decreases with higher drive strength, which is expected behavior.

3.2 Sample Measured Values

At:

$$t_{in} = 0.1225ns, \quad C_L = 0.0094pF$$

For invx1:

- cell rise = $6.66 \times 10^{-10}s$
- cell fall = $1.52 \times 10^{-10}s$
- rise transition = $5.24 \times 10^{-10}s$
- fall transition = $1.09 \times 10^{-10}s$

Chapter 4

RC vs SPICE Comparison

Using first-order RC model:

$$t_p = 0.69RC$$

From Tutorial 2:

- Driver resistance $R \approx 10k\Omega$
- Inverter capacitance $C_{inv} \approx 1.597 \times 10^{-15}F$

$$C_L = 0.0094pF = 9.4 \times 10^{-15}F$$

$$t_p = 0.69 \times 10^4 \times 9.4 \times 10^{-15} = 6.486 \times 10^{-11}s$$

Comparison:

- RC delay: $6.49 \times 10^{-11}s$
- SPICE delay: $6.66 \times 10^{-10}s$

Observation:
SPICE delay is about 10× larger than RC estimate.
Explanation:
The RC model assumes linear resistance and lumped capacitance. However, real CMOS behavior is nonlinear due to:

- transistor switching behavior
- parasitic capacitances
- input slew effects
- different pull-up and pull-down strengths

Therefore, RC is only an approximation, while SPICE provides accurate results.

Chapter 5

Discussion

The results show expected trends:

- Delay increases with load capacitance
- Stronger cells reduce delay
- Slower input transitions increase delay

The inverter family clearly demonstrates drive strength scaling.

Chapter 6

Conclusion

A full standard cell library was successfully designed and characterized using SKY130 and NGSPICE. Automated scripts enabled generation of complete NLDM tables for all cells.

The comparison between RC and SPICE models highlights the importance of transistor-level simulation for accurate timing analysis.